DIAMOND'S NEXT JOB: Keeping Chips Cool

With AI processors and GPUs packing ever more power into ever smaller areas, heat is becoming the limiting factor in chip design. A material best known for engagement rings is emerging as an unlikely answer.

NIDHI AGARWAL

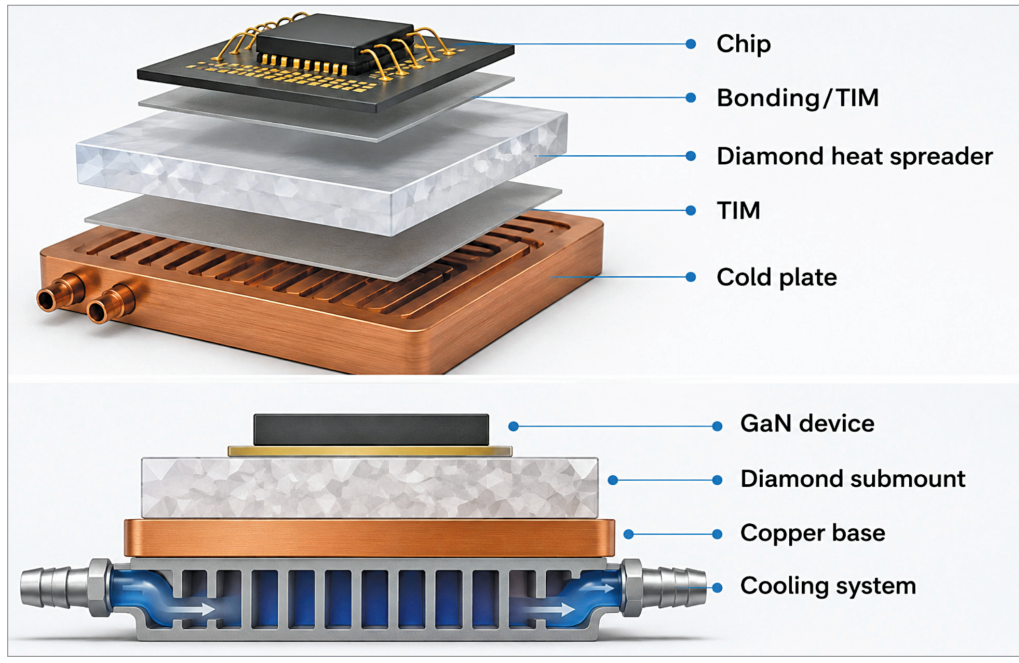
Diamonds have long been associated with jewellery. But what if the same material could also help keep a high-power chip from overheating? That is where diamond is finding a new role in electronics. As chips shrink and grow more powerful, they concentrate more heat in smaller areas. This thermal challenge impacts artificial intelligence (AI) processors, graphics processing units (GPUs), data-centre hardware, lasers, photonics systems, and aerospace electronics. Diamond provides an ideal solution by rapidly drawing heat away from these critical hot spots.

Five times better than copper

The secret lies in thermal conductivity. A single-crystal diamond can reach around 2200 watts per metre-Kelvin, more than five times the roughly 400 W/m·K offered by copper. At the same time, diamond is an electrical insulator, so it can manage heat without creating an unwanted electrical connection.

This matters most where chips develop localised hot spots. Rather than letting heat build up in one place, a diamond heat spreader carries it away and lowers the chip's junction temperature, giving GPUs and other high-power processors more headroom to stay within their thermal limits, and in some designs, easing the load on the cooling system.

So, why are we seeing more interest in diamonds now? One reason is that producing them for industrial use has become more practical. Lab-grown diamonds were initially developed largely for jewellery, but chemical vapour deposition (CVD), and now microwave plasma CVD (MPCVD), can grow diamonds specifically for applications such



Where diamond fits inside electronic packages (Credit: ChatGPT)